High-Speed Serial PCB Design

PCIe • USB 3.x / 4 • SATA • Thunderbolt • DisplayPort

Gbps-class differential signalling — where SI gets serious

Audience: Intermediate / Advanced Engineering Students

Topology • Vias • AC coupling • Eye margin • Stackup • Materials

Why High-Speed Serial is Different

GHz signalling demands a new mindset

Bit period

PCIe Gen 5: UI = 31 ps
Sets all timing margins.

Loss budget

Insertion loss budgets:
PCIe Gen 5: ~30 dB total

Jitter

Random + deterministic
Limits eye opening.

Reflections

Even small Z_0 bumps
appear as eye closure.

Materials

FR-4 caps at ~10 Gbps
Low-Df materials needed beyond.

Verification

Eye diagrams + TDR mandatory.
Sim before fab.

High-Speed Serial Standards Landscape

Speed + impedance + materials by standard

Standard	Data rate	Diff Z_0	Material	Max trace
USB 2.0 (HS)	480 Mbps	90 Ω	Standard FR-4	~3 m cable
USB 3.0 (SS, 5 Gbps)	5 Gbps	90 Ω	Standard FR-4	~150 mm
USB 3.2 (Gen 2, 10 Gbps)	10 Gbps	90 Ω	Mid-loss FR-4	~125 mm
USB4 (20-40 Gbps)	20-40 Gbps	85 Ω	Low-loss FR-4 / Megtron	~75-100 mm
PCIe Gen 1-2	2.5-5 Gbps	85 Ω	Standard FR-4	~250 mm
PCIe Gen 3	8 Gbps	85 Ω	Low-loss FR-4	~125 mm
PCIe Gen 4-5	16-32 Gbps	85 Ω	Megtron 6 / Tachyon	~75 mm
SATA III	6 Gbps	100 Ω	Standard FR-4	~150 mm
Thunderbolt 4	40 Gbps	85 Ω	Low-loss FR-4	~75-100 mm
DisplayPort 2.1	20-80 Gbps	100 Ω	Low-loss FR-4	Active cable

Topology for High-Speed Serial

Point-to-point + AC coupling = the standard pattern

- Always point-to-point — no daisy-chain, no stubs
- AC coupling: series caps (0.1-1 µF) between driver and trace
- Caps protect against DC ground bounce, allow different common modes
- Common cap size: 0402 X7R 100 nF (sometimes 0201 for very-high-speed)
- Place AC coupling caps NEAR the driver (transmitter side)
- TX (driver) side has AC coupling; RX (receiver) side typically does not
- Per-direction caps: TX D+ → cap → wire to RX D+; same for D−
- Cap location: in line with the trace, no stub branches

Stackup for High-Speed Serial

What works at each speed grade

- USB 3.0 (5 Gbps): 4-layer FR-4 is fine if traces < 100 mm
- USB 3.2 Gen 2 (10 Gbps): 4-layer with low-loss FR-4 (Df < 0.012)
- USB4 / PCIe Gen 4 (16 Gbps+): 6-8 layer with mid-loss Megtron 4 / Rogers RO4350
- PCIe Gen 5 / Thunderbolt 4 (32 Gbps+): 8-10 layer Megtron 6 or Tachyon
- Reference plane: GND directly adjacent to every high-speed signal layer
- Stripline (sandwiched between two GND planes) gives best SI; microstrip is OK shorter runs
- Inner-layer routing reduces radiation but slows propagation (higher εr)
- Hybrid stackups (FR-4 outer + low-loss inner) optimize cost

Insertion Loss Budget

How much trace your spec allows

- Total loss budget = trace + via + connector + cable losses
- PCIe Gen 5 total budget: ~36 dB at Nyquist (16 GHz)
- Standard FR-4 loss: ~0.6 dB/inch @ 5 GHz, ~1.5 dB/inch @ 16 GHz
- Low-loss FR-4 (Df 0.005-0.008): ~0.4 dB/inch @ 5 GHz
- Megtron 6 (Df 0.002): ~0.2 dB/inch @ 5 GHz
- Connector loss budget: 1-2 dB per connector (PCIe / USB-C)
- Cable loss: 0.5-2 dB per metre (longer = lossier)
- Plan budget WORK BACKWARDS: total - connectors - cable = max trace loss

Via Design for High-Speed Serial

Every via is a reflection

- Minimize via count on high-speed signals (each via = ~0.5 dB at 10 GHz)
- Back-drilling: drill OUT the unused via stub from the bottom (controlled depth)
- Stub length < $\lambda/10$ of Nyquist frequency to avoid resonance
- Anti-pad size: 40-60 mil for PCIe Gen 3+, larger = closer to 85 Ω impedance
- Via-in-pad (filled + plated over): used in BGAs to enable short escape routing
- Microvias (laser-drilled, blind, 75-150 µm): minimum inductance, used at GHz speeds
- Stitching vias adjacent to signal vias (within 1-3 mm) for return current
- HDI stackups enable shortest, lowest-stub vias

Eye Diagrams — How Quality is Measured

The universal high-speed serial quality metric

- Eye diagram = overlay of many bit transitions on the scope
- Wide-open 'eye' = clean signal; closed eye = SI failure
- Measurements: eye height (vertical opening), eye width (horizontal)
- Receiver compliance: defined eye mask the signal must clear
- PCIe Gen 3 mask: 175 ps × 200 mV opening minimum
- USB 3.2 mask: defined in spec section 6.7

How to capture

Use a high-bandwidth scope (≥ 4× UI rate) with a high-speed probe.
Trigger on data clock, accumulate many UIs.
Overlay → eye.

Pre-fab simulation

HyperLynx / ADS / IBIS-AMI models predict the eye BEFORE fab.
Driver/receiver IBIS files include equalization.
Run pre-layout for topology check; post-layout for verification.

Equalization (EQ)

Compensating for trace loss

- Pre-emphasis at TX boosts high-frequency content of signal
- De-emphasis: lowers low-frequency content (similar effect)
- Receiver continuous-time linear EQ (CTLE) boosts attenuated frequencies
- Decision feedback EQ (DFE) cancels post-cursor inter-symbol interference
- Adaptive EQ: link partners negotiate optimal settings during training
- Most modern serdes (PCIe Gen 3+, USB 3.2+) have built-in EQ
- Your PCB job: keep trace loss WITHIN the budget the EQ can recover
- Over-budget = link drops or auto-negotiates to lower speed

USB-C Specifics

Modern combo connector — multiple protocols on one port

- USB-C connector carries: USB 2.0 + USB 3.x + DisplayPort + Thunderbolt + Power Delivery
- Per-side TX/RX diff pairs (allows reversal) — duplicated nets
- CC1 / CC2 pins: orientation detection + PD negotiation
- SBU1 / SBU2: sideband channel (DisplayPort AUX, etc.)
- VBUS: up to 240 W (PD 3.1 EPR at 48 V × 5 A)
- Add common-mode choke per diff pair for EMI compliance
- ESD: TVS array on all data lines (low-cap for SS pairs)
- PD controller IC: TPS25750, CH224K, FUSB302 — handles negotiation

High-Speed Serial Layout Rules

The non-negotiable checklist

- Controlled-impedance throughout — fab quote must spec ±10% (or tighter)
- Diff pair routed together on same layer; symmetric escape from connectors
- Length match within pair: < 5 mil for USB 3+, < 2 mil for PCIe Gen 3+
- Continuous reference plane (GND) under every high-speed signal
- AC coupling caps at TX side, in-line, no stubs
- Connector AC coupling for cable-side (separate from on-board caps)
- ESD/EMI components AFTER coupling caps (TVS, CMC)
- Vias: minimum count; stub-controlled via back-drilling; stitching nearby

Common High-Speed Serial Mistakes

What link-training failure means

- Trace exceeds loss budget → link auto-negotiates down or fails
- Long via stubs at high speed → resonance in band → bit errors
- Plane split crossed by diff pair → return path detour → reduced eye
- AC coupling cap missing or wrong value → DC ground bounce → flaky link
- Reference plane discontinuous → impedance jumps mid-trace
- USB 3.x SS pairs on FR-4 with > 150 mm length → high loss, fails compliance
- Length match between PAIRS forgotten (only within-pair matched) → byte-lane skew
- ESD device added with high capacitance (> 1 pF) → kills GHz signals

Verification Workflow

From sim to silicon validation

Pre-layout sim

HyperLynx / ADS
Validate topology + budget

Post-layout sim

Extract PCB parasitics → simulate eye + losses

TDR measurement

Measure actual Z_0
Find discontinuities

VNA measurement

S-parameter sweeps
Insertion + return loss

Eye-diagram capture

Scope + high-BW probe
Looks-good-or-not check

Compliance test

Lab equipment + golden test fixture
USB-IF, PCI-SIG certify

Best Practices

Habits for production GHz designs

- Pick stackup + material EARLY (before schematic)
- Use vendor reference design + layout guide
- Minimize trace length (shorter = less loss)
- AC coupling caps at TX side only
- Stub-free vias (back-drill or microvia)
- Pre-layout SI sim — validate topology

- Continuous GND reference
- Tight length match within pair
- Pair-to-pair spacing ≥ 4× within-pair gap
- Post-layout sim with extracted parasitics
- Low-cap TVS for ESD
- TDR + eye verify on prototype

Key Takeaways

Key takeaways from this lecture

Loss budget rules

Plan trace, via, connector, cable.

Stackup + material

FR-4 caps at ~10 Gbps; low-loss past it.

Back-drill stubs

Resonances kill at GHz speeds.

AC coupling at TX

DC isolation between transmitter and receiver.

Sim → measure

Pre-layout sim, post-fab TDR + eye.

USB-C is many protocols

Plan for all you support.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- PCI-SIG specifications (free) — PCIe Base + Card Electromechanical
- USB-IF specifications — USB 3.x, USB4, USB-PD
- Eric Bogatin — Signal Integrity Simplified
- Stephen Hall — Advanced Signal Integrity
- HyperLynx / Sigrity SI tools (Mentor / Cadence)
- Megtron / Tachyon / Rogers materials — datasheets
- Phil's Lab / Robert Feranec YouTube — practical SI walkthroughs
- DesignCon papers — annual high-speed PCB conference

Thank you • Build something this week.